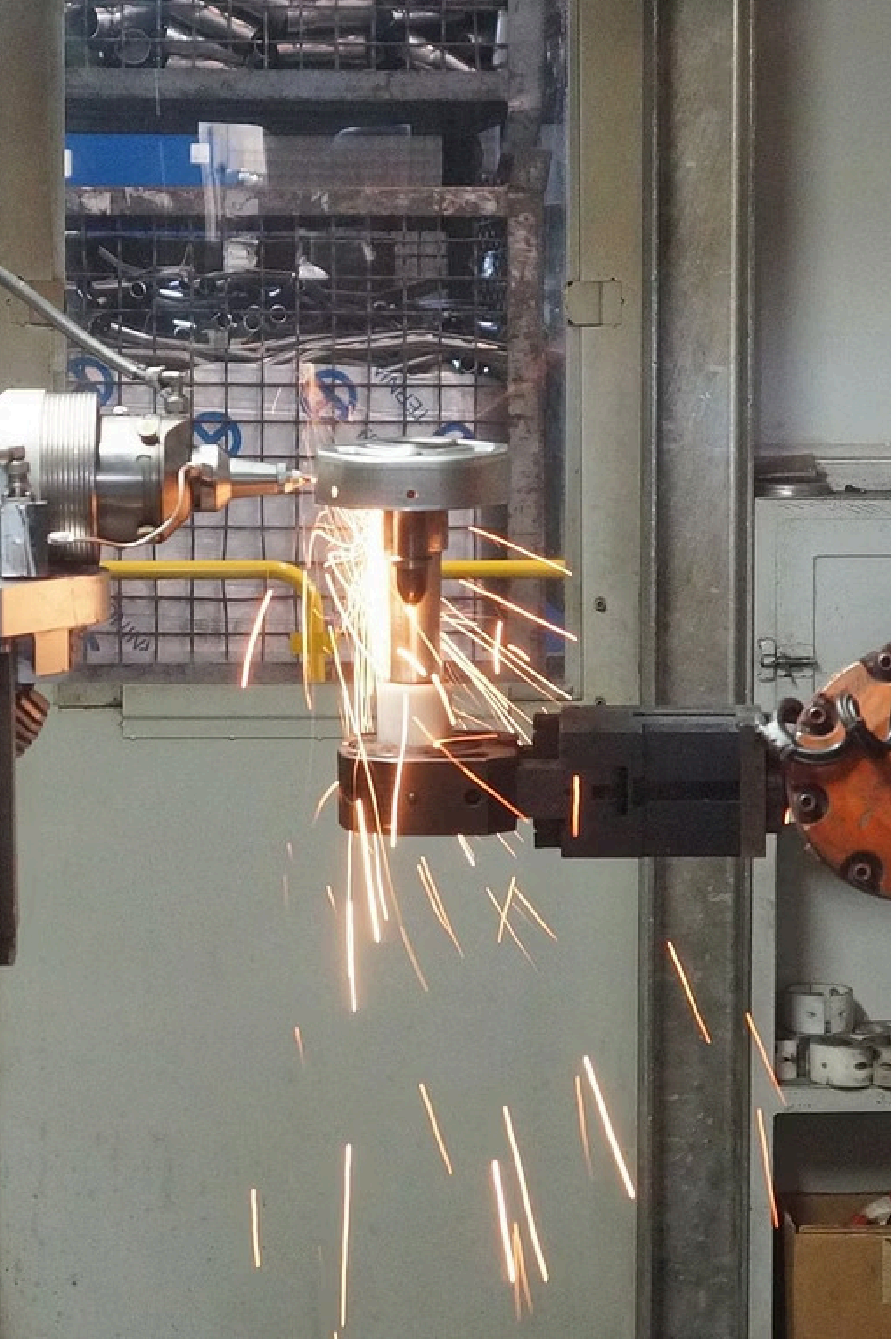


Safety Communication Protocols in Industrial Automation

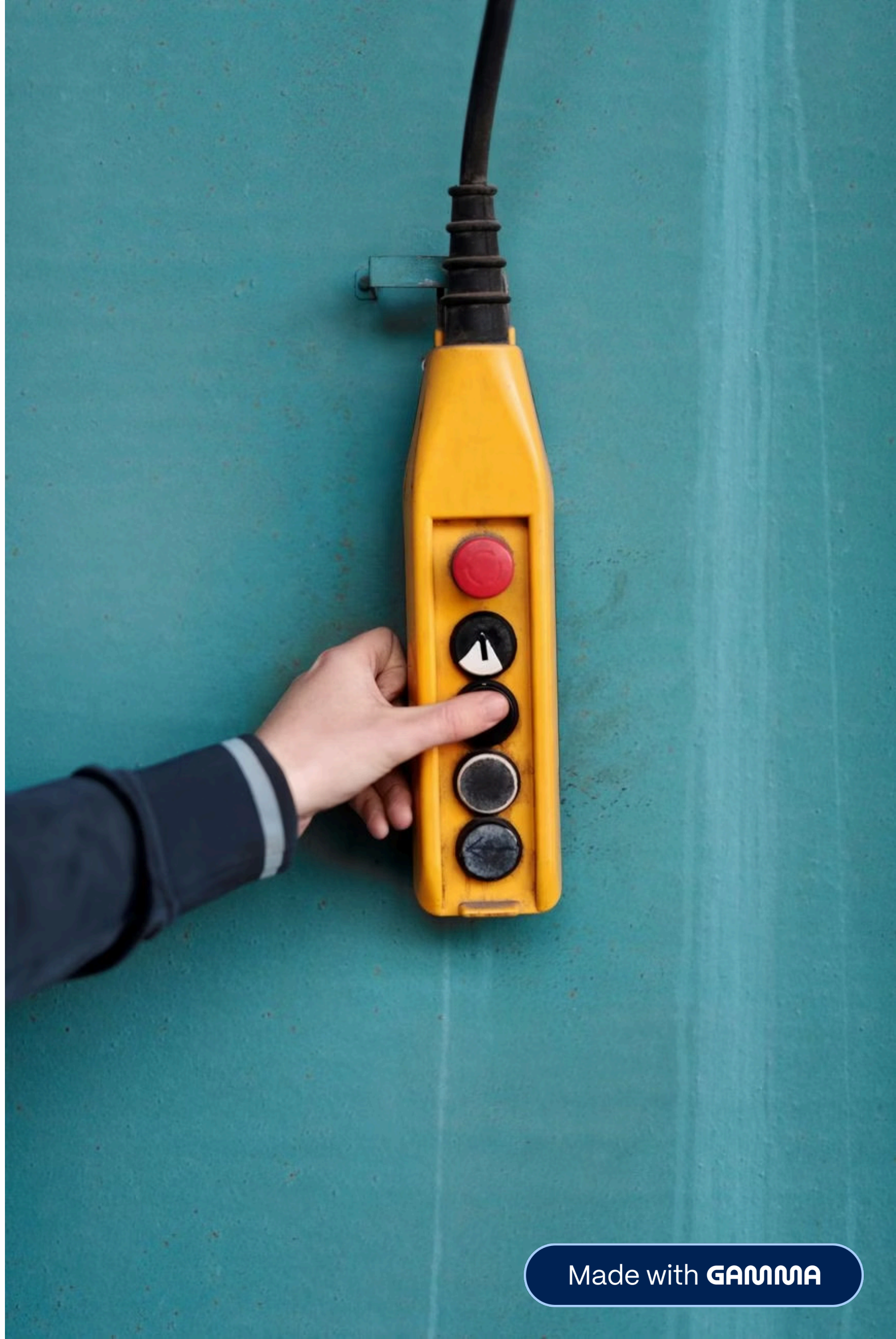
A Comparative Study of PROFIsafe, FSoE, CIP Safety, openSAFETY, and CC-Link IE Safety

HOCHSCHULE HAMM-LIPPSTADT · ELECTRONIC ENGINEERING · ICS TEAM A2

Would You Use Wi-Fi to Stop a 200-Tonne Press?

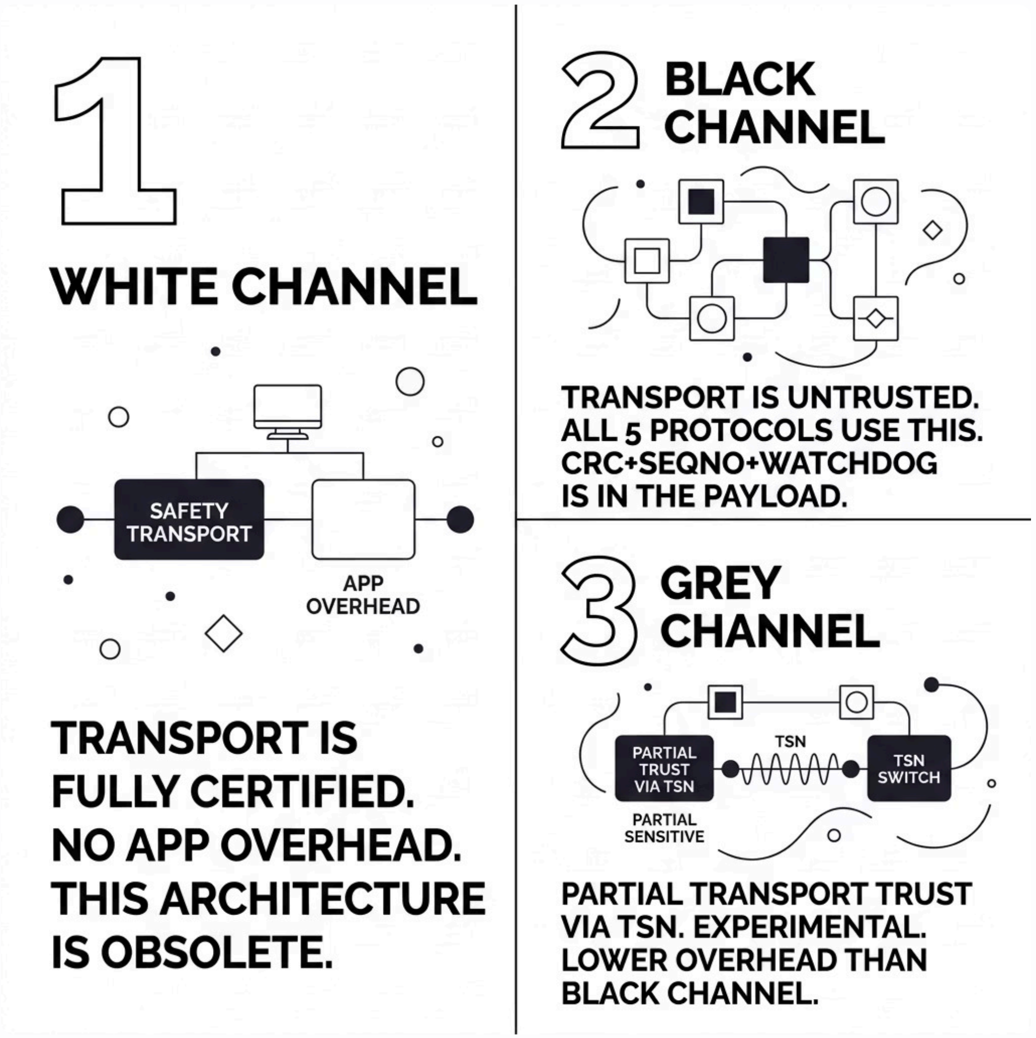
Standard Ethernet and Wi-Fi offer **best-effort delivery** — no guarantee any frame arrives within a bounded time. For a safety circuit, a single late or lost packet can mean a robot arm strikes an operator or a press closes on a hand.

> ⚠ **Functional Safety (IEC 61508):** SIL 3 requires PFH < 10^{-7}/h. Certified communication links are mandatory for Category 3 / PL d and above.



The Black Channel Principle

All five protocols are governed by **IEC 61784-3** and treat the transport layer as **fully untrusted**. Error protection is embedded directly in each safety payload.



What the Black Channel Detects

All **7 transmission error classes** defined by IEC 61784-3:

- Corruption, unintended repetition, incorrect sequence
- Loss, unacceptable delay, insertion, masquerading

All five protocols—PROFIsafe, FSoE, CIP Safety, openSAFETY, CC-Link IE Safety—use the Black Channel model.

PROFIsafe (CP 3/1) & FSoE (CP 12/1)

PROFIsafe

F-Host/F-Device model. F-PDU with **CRC-24**, 8-bit F-Address, consecutive number. Payload up to **123 bytes**. Safety overhead ~4 B. Min. cycle ~1 ms, jitter <1 μs (IRT). **35M+ certified nodes**—dominant in EU automotive (VW, BMW) and process industry (BASF).

FSoE

Master/Slave model. **CRC-8/16**, 2-byte Safety Header Block. Safety overhead ~6 B. **Autonomous slave passivation** on watchdog expiry—no master round-trip. Min. cycle **~125 μs**, jitter <1 μs (best in class). Preferred for high-axis motion control (FANUC, TRUMPF, Beckhoff).

CIP Safety (CP 2/1) & openSAFETY (CP 13/1)

CIP Safety

Originator/Target over EtherNet/IP. **Dual-packet format**: process data + bitwise complement transmitted together—robust against systematic failures. Safety overhead **~12 B** (highest). Min. cycle ~1–10 ms, jitter ~100 µs. De-facto standard in **North America** (GM, Boeing, P&G). No gateway needed for mixed safety/standard networks.

openSAFETY

Frame-in-frame over POWERLINK/Modbus TCP. Two independent SPDO sub-frames, each with own CRC (CRC-8 + CRC-16). Passivation only when **both** fail—reduces nuisance trips. Overhead ~11 B, min. cycle ~400 µs. **Only royalty-free, fully open spec**—any vendor may certify independently. Favoured by EU OEMs (Engel, Krones, B&R)).

CC-Link IE Safety (CP 8/1)

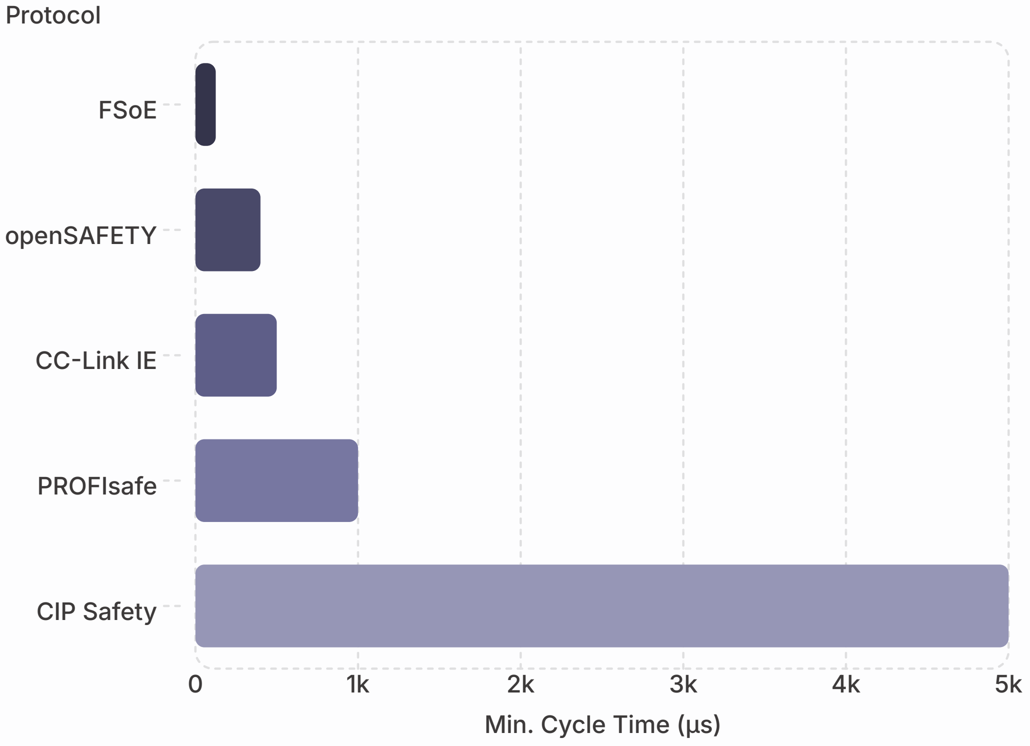
Architecture

Safety Communication Layer (SCL) mapped into CC-Link IE Field cyclic frames. **CCITT CRC-16** + node address + frame counter for anti-replay. Safety overhead ~6 B. Min. cycle ~500 µs.

Key Constraints

- Topology limited to **ring and line** only
- Centralised passivation via **Mitsubishi MELSEC iQ-R** safety PLC
- Certified under **JIS B 9961** (internationally recognised)
- Effectively **mandatory** for Tier 1/2 suppliers to Japanese OEMs (Toyota, Denso, Panasonic)

Performance Comparison



Key Takeaways

- **FSoE** leads with ~125 µs cycle and <1 µs jitter — only viable choice for servo control <2 ms
- **openSAFETY** follows at ~400 µs via POWERLINK isochronous scheduling
- **CIP Safety** accepts higher jitter (~100 µs) for EtherNet/IP interoperability
- All five achieve **SIL 3 / PL e** — PFH < 10^{-8}/h

For E-stop and light-curtain applications (safety times >10 ms), all five protocols provide adequate headroom.

Protocol Architecture Comparison

Feature	PROFIsafe	FSoE	CIP Safety	openSAFETY	CC-Link IE
CPF	3/1	12/1	2/1	13/1	8/1
Channel	Black	Black	Black	Black	Black
Max Payload	123 B	126 B	Variable	8 B (SPDO)	8 B
CRC	24-bit	8/16-bit	16-bit (S1)	8+16-bit	16-bit
Safety Overhead	~4 B	~6 B	~12 B	~11 B	~6 B
Topology	Any	Any	Any	Any	Ring/Line
Open Spec.	No	No	No	**Yes**	No

All five achieve SIL 3 / PL e. Differentiation lies in cycle time, topology, overhead, and ecosystem openness.

Industry Deployment by Sector

Sector	PROFIsafe	FSoE	CIP Safety	openSAFETY	CC-Link IE
Automotive EU	**Primary** (VW, BMW)	Common	Common	—	—
Automotive NA	Common	—	**Primary** (GM, Ford)	—	—
Automotive APAC	Common	Common	—	—	**Primary** (Toyota)
Process Industry	**Primary** (BASF)	—	Common	—	—
Machine Building	**Primary**	Common	—	Common (Engel)	—
Food & Beverage	Common	Common (Tetra Pak)	Common	Common	—
Packaging	Common	**Primary** (Bosch)	Common	**Primary** (Krones)	—
Electronics	Common	**Primary**	Common	—	Common (Panasonic)

No protocol dominates all sectors. PROFIsafe leads in Europe/process; CIP Safety in North America; CC-Link IE in APAC automotive. FSoE and openSAFETY are application-driven, not regionally constrained.

Same Hardware, Different Protocol

The protocol is **not** determined by hardware — it's chosen by the system integrator based on regional infrastructure and supply chains.

🇪🇺 European Line

Siemens SIMATIC S7-1500F runs **PROFIsafe** over PROFINET

Volkswagen, BMW, BASF, ThyssenKrupp

🇺🇸 North American Line

Identical hardware runs **CIP Safety** over EtherNet/IP

General Motors, Procter & Gamble, Boeing

A safety engineer working across sites must be competent in multiple protocols simultaneously.

Conclusions & Selection Guidance

PROFIsafe

Largest installed base (35M+ nodes). Best for EU automotive & process industry with Siemens infrastructure.

FSoE

Fastest cycle (125 μs) & lowest jitter. Only choice for <2 ms motion-control reaction time.

CIP Safety

De-facto in North America. Highest overhead, but seamless EtherNet/IP integration—no gateway needed.

openSAFETY

Only royalty-free open spec. Lowest vendor entry barrier. Favoured by EU machine builders.

CC-Link IE

Mandatory in APAC/Japanese automotive supply chain. Ring/line topology only.

Three selection factors: (1) Transport already deployed at site · (2) Regional device ecosystem · (3) Required safety reaction time. All five reach SIL 3 / PL e—the ceiling is not a differentiator.